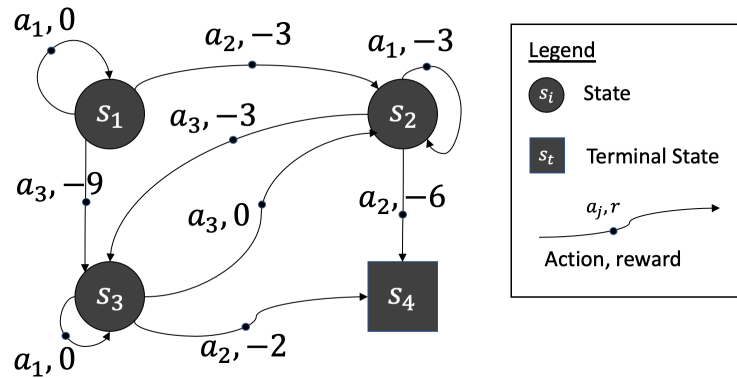


Exercise A

Let's consider the generic Markov Decision Process (MPD) $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ and a generic policy π .

- Provide the pseudo-code for the Iterative Policy Evaluation algorithm to estimate V_π in the generic MDP; in the algorithm consider
 - having reached K iterations as unique exit condition;
 - the 'in-place' version of the algorithm.
- Consider the deterministic MDP reported in the Figure above and consider a uniform random policy π , ie:

$$\pi(a_{i1}, s_j) = \pi(a_{i2}, s_j) \quad \forall a_{i1}, a_{i2} \in \mathcal{A}, \forall s_j \in \mathcal{S}$$

Estimate V_π by using iterative policy evaluation with:

- $\gamma = 1$;
- initial estimation $V_\pi = (0, 0, 0, 0)$;
- consider the 'in-place' version of iterative policy evaluation;
- stop the algorithm after reaching $K = 3$ as exit condition.

Report the procedure with calculations [hint: after 3 iterations, the estimation should be $V_\pi = (-\frac{86}{9}, -\frac{64}{9}, -\frac{112}{27}, 0)$].

- Report the pseudo-code of the Policy Iteration algorithm (using iterative policy evaluation): for the Policy Evaluation step, just refer to the algorithm reported at point 1.
- Consider the 'true' $V_\pi = (-13, -\frac{26}{3}, -\frac{16}{3}, 0)$: by applying one step of policy improvement derive a new deterministic policy π' .

Exercise B

- Provide the pseudo-code for the Monte Carlo prediction algorithm to estimate V_π in starting from the availability of trajectories (or the capability to generate trajectories); report the 'first-visit' version of the algorithm.
- Let's consider the following trajectories $(s, a, r, s, a, \dots)$ obtained in the MDP reported in Exercise 1 by applying policy π (as defined in Exercise 1):
 - Episode #1: $s_1, a_3, -9, s_3, a_3, 0, s_2, a_1, -3, s_2, a_3, -3, s_3, a_2, -2, s_4$
 - Episode #2: $s_1, a_1, 0, s_1, a_1, 0, s_1, a_2, -3, s_2, a_3, -3, s_3, a_3, 0, s_2, a_1, -3, s_2, a_3, -3, s_3, a_3, 0, s_2, a_2, -6, s_4$;
 - Episode #3: $s_1, a_3, -9, s_3, a_1, 0, s_3, a_2, -2, s_4$;
 - Episode #4: $s_1, a_3, -9, s_3, a_3, 0, s_2, a_3, -3, s_3, a_2, -2, s_4$;
 - Episode #5: $s_1, a_1, 0, s_1, a_3, -9, s_3, a_3, 0, s_2, a_3, -3, s_3, a_3, 0, s_2, a_3, -3, s_3, a_2, -2, s_4$.

Let's consider the full Reinforcement Learning scenario, ie. the MDP is not available. Exploiting the Monte Carlo prediction algorithm, estimate V_π by considering: (I) $\gamma = 1$; (II) initial estimation $V_\pi = (0, 0, 0, 0)$.

- If we want to estimate Q_π instead of V_π , how the algorithm described in point 1 has to be changed?
- Estimate Q_π from the trajectories in point 2. Consider again:
 - $\gamma = 1$;
 - initial estimation $Q_\pi(s, a) = 0 \forall s \in \mathcal{S}, \forall a \in \mathcal{A}$.

- How can we derive a new policy π' based on the information collected in point 2 and point 4?

REPORT YOUR ANSWERS TO EXERCISE A AND EXERCISE B IN THE PROTOCOL SHEET

Question 1

When dealing with a large MDP, the value function $v_\pi(s)$ is estimated by an function approximation $\hat{v}(s, \mathbf{w})$, that is,

$$v_\pi(s) \approx \hat{v}(s, \mathbf{w}),$$

where $\mathbf{w}$ is a parameter vector. Typically the size of $\mathbf{w}$, denoted by n is much smaller than the number of states of the MDP.

1. Assume that, thanks to an oracle, the exact value of $v_\pi(s)$ is known for any s . Describe how to estimate the vector $\mathbf{w}$ adopting a stochastic gradient descent algorithm (*Introduce a suitable function cost to be minimized...*).
2. Consider the linear case, that is, $\hat{v}(s, \mathbf{w}) = \mathbf{w}^T \mathbf{x}(s)$ where $\mathbf{x}(s)$ is a feature vector. Under which conditions on the step-size sequence, the algorithm is guaranteed to converge to a global optimum of the functional cost introduced in the previous question? Is it possible to converge if a constant step-size is adopted?
3. Assume now that no knowledge of the true value function $v_\pi(s)$ is available. Describe a Gradient Monte-Carlo algorithm for estimating $\hat{v} \approx v_\pi(s)$. Consider the linear case; is the algorithm still converging to the optimal solution? If so, under which conditions on the step-size sequence?
4. Describe the semi-gradient $TD(0)$ algorithm for estimating $\hat{v} \approx v_\pi(s)$. Address the following questions
 - (a) Is this algorithm a true stochastic gradient algorithm? Why?
 - (b) Consider the linear case, i.e., $\hat{v}(s, \mathbf{w}) = \mathbf{w}^T \mathbf{x}(s)$. Do you expect the algorithm to converge to the optimal solution?
 - (c) Consider the continuing case $\gamma < 1$ (γ is the discounting factor) and consider again the linear case. Is it possible to guarantee the convergence of the algorithm to its fixed point adopting a constant step-size? Is it possible to characterize the distance of the fixed point from the optimal solution?
5. In general, what are the main pros and cons of adopting a bootstrapping method instead of a Monte-Carlo approach to estimate $\hat{v} \approx v_\pi(s)$?

REPORT YOUR ANSWER TO QUESTION 1 IN THE FOLLOWING AND IN THE NEXT PAGES

A full-page sheet of white graph paper with a light gray grid. The grid consists of small squares, approximately 10 units wide by 10 units high. There are no margins or other markings on the page.

This image shows a full page of blank graph paper. The grid consists of small, uniform squares formed by thin, light gray lines. There are no margins, text, or other markings on the page.

This image shows a full page of blank graph paper. The grid consists of thin, light gray horizontal and vertical lines that intersect to form small squares across the entire surface. There are no margins, text, or other markings on the paper.